August 25, 2026

The Editor-in-Chief
Acta Astronautica
International Academy of Astronautics

Subject: Submission of original research article — *"Eclipse-gated transient energy gain for passive solar sails in near-polar low Earth orbit"*

Dear Editor,

I am pleased to submit my manuscript, *"Eclipse-gated transient energy gain for passive solar sails in near-polar low Earth orbit"*, for consideration as an original research article in *Acta Astronautica*. The work falls within the journal's scope of astrodynamics and space-borne system design, and addresses a question of direct relevance to passively oriented small solar-sail spacecraft in low Earth orbit (LEO).

Aim and main findings. The paper investigates whether a passively oriented, spin-stabilised solar sail in near-polar LEO can exhibit a bounded interval of net orbital-energy gain through eclipse-modulated solar-radiation-pressure (SRP) forcing. Combining a reduced analytical model with three-dimensional numerical propagations, I show that the regime exists, but only within a narrow LTDN–pitch envelope: at 700 km, only 89 of 900 sampled cases ($\sim 10\%$) show positive semi-major-axis excursion within a 14-day window, with peak gain $\Delta a \approx +20.07$ km. Eclipse is causally decisive — the eclipse-on case rises by $+10.31$ km over 14 days while the eclipse-off counterpart falls by -47.75 km — and the effect is transient and self-limiting through coupled apogee growth, eccentricity build-up, and perigee erosion.

Independent validation and open toolchain. The paper also pursues a methodological goal. I re-implement the model in an independent open-source propagator, which reproduces the gain lobe, the eclipse-driven sign reversal, and the transient–breakdown signature, confirming that the mechanism is not an artefact of a single implementation. The propagator is cross-validated component by component against two independent references: Ansys STK with the Astrogator module, and the open-source Orekit library. Two-body and J_2 dynamics agree with STK to better than 0.05 km over 14 days, whereas the drag-induced decay differs by close to an order of magnitude despite matching instantaneous force; Orekit reproduces the open-source decay to 4×10^{-10} km, localising that discrepancy to STK's drag application rather than to the open-source implementation. Sensitivity testing identifies the choice of atmospheric density model, rather than the SRP mechanism or the drag conventions, as the dominant uncertainty governing where the regime is realisable. The full toolchain and data are released openly.

Fit, novelty, and implications. *Acta Astronautica* publishes work in astrodynamics, space transportation, and spacecraft system design — the natural home for this contribution. To my knowledge, the eclipse-modulated transient energy-pumping regime has not previously been isolated and characterised in this form for spin-stabilised sails in LEO. Prior literature emphasises long-term or equilibrium-like SRP–drag balance; I instead show that eclipse timing alone, without active control, can flip the sign of the orbit-averaged SRP work integral over a bounded horizon, while also exposing the coupled mechanisms that close the regime. The result clarifies a real but narrow opportunity for passive orbit shaping and provides a cautionary example of how the same dynamics that enable early gain drive its eventual breakdown.

Submission status. The manuscript is original, has not been published elsewhere, and is not under consideration by any other journal. All listed authors have read and approved the submitted version. We have no prior or concurrent submissions related to this work.

We thank you for considering our manuscript and look forward to the review process.

Yours sincerely,

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